CSE 4105 · SYSTEM PROGRAMMING · SHORT COURSE

Class 2

Machine Language and the SIC Machine

instructions, formats and addressing, seen through C

CLASS

2 / 10

19 slides, taken from the full course:
Lecture 2

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One C statement, three levels of language

C **High-level language**

```c
beta = alpha + incr - 1;
```

One line can hide many steps. The compiler works out the order and where the values live.

SIC **Assembly language**

```
LDA   ALPHA    ; A ← alpha
ADD   INCR     ; A ← A + incr
SUB   ONE      ; A ← A - 1
STA   BETA     ; beta ← A
```

One line = **one machine instruction**. Names (mnemonics, labels) instead of numbers.

hex **Machine language**

```
00 100F
18 1012
1C 1015
0C 1018
```

What is really stored in memory. The first byte is the **opcode**, the rest is the **address**.

Compiler = C → assembly (many-to-one, clever). **Assembler** = assembly → machine code (one-to-one, mechanical). The assembler is what we build in Lecture 3.

Book: Beck §1.3.1, Appendix A (opcodes)

SIC at a glance: memory, registers and data

Memory
32,768 bytes (2^{15}), byte-addressed. A **word is 3 bytes**, and the address of a word is the address of its lowest byte.

Registers
Five, each one word (24 bits): **A** accumulator, **X** index, **L** linkage, **PC** program counter, **SW** status word (holds the condition code).

Data
Integers: 24-bit two's complement. **Characters**: 8-bit ASCII. **No floating point** in plain SIC.

C int vs SIC word

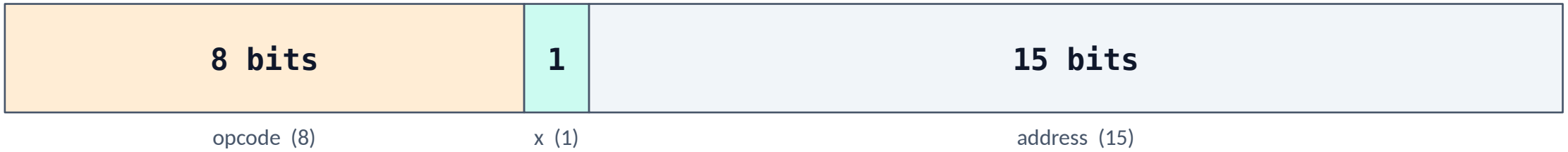
	C int (lab PC)	SIC word
Size	32 bits	24 bits
Smallest	−2,147,483,648	−8,388,608
Largest	2,147,483,647	8,388,607
−9 is stored as	FF FF FF F7	FF FF F7

Same idea, two's complement; only the width differs.

Like C: an `int` is a fixed number of bytes, and negative numbers use two's complement. **Unlike C:** there is no `char` type. A character is just one byte, and only the instructions LDCH and STCH treat it as one.

Book: Beck §1.3.1

The SIC instruction format: 24 bits, three fields



Every SIC instruction is exactly **3 bytes**. The **opcode** says which operation. The **x bit** says whether to use the index register. The **address** says where the operand is in memory: 15 bits, which is exactly enough for 32,768 bytes.

Direct addressing (x = 0)

Target address: `TA = address`

SIC: `LDA ALPHA` **C idea:** `alpha`

The operand is at the address written in the instruction.

Indexed addressing (x = 1)

Target address: `TA = address + (X)`

SIC: `LDCH STR1,X` **C idea:** `str1[i]`

Add the contents of X first. X plays the role of the array index i.

Book: Beck §1.3.1 (instruction format, addressing modes)

Worked example: encode STCH BUFFER,X by hand

Given: BUFFER is at address **1039** (hex), and the opcode of STCH is **54** (Beck, Appendix A).

1	**Opcode in binary**	`54 hex → 0101 0100`	8 bits
2	**Indexed, so x = 1**	`the operand ends in ,X → x = 1`	1 bit
3	**Address in 15 bits**	`1039 hex → 001 0000 0011 1001`	15 bits
4	**Join the 24 bits**	`0101 0100 | 1 | 001 0000 0011 1001`	24 bits
5	**Regroup in fours**	`0101 0100 1001 0000 0011 1001`	
6	**Read as hex**	`5 4 9 0 3 9 → 549039`	

Shortcut

Because x is the top bit of the address field, setting x = 1 just **adds 8000 hex** to the address:

`1039 + 8000 = 9039`

So the answer is opcode `54` followed by `9039`.

Try it: LDA ALPHA with ALPHA at 1033 and LDA = 00. → x = 0, so the answer is simply `001033`.

Book: Beck §1.3.1; the same instruction appears in Beck's Fig. 2.2

The SIC instruction set, grouped by job

Group	Instructions	Opcodes (hex)	What they do	C idea
Load / store	`LDA m · STA m`	`00 · 0C`	A ← (m) · m ← (A)	`x = y`
	`LDX · STX · LDL · STL`	`04 · 10 · 08 · 14`	same for X and L	
	`LDCH m · STCH m`	`50 · 54`	one byte ↔ rightmost byte of A	`c = s[i]`
Arithmetic	`ADD · SUB · MUL · DIV`	`18 · 1C · 20 · 24`	A ← A op (m)	`+ − * /`
Compare	`COMP m`	`28`	CC ← compare A with (m)	`a > b ?`
	`TIX m`	`2C`	X ← X + 1, then compare X with (m)	`i++; i < n ?`
Jump	`J · JEQ · JGT · JLT`	`3C · 30 · 34 · 38`	PC ← m (always / if CC is = > <)	`goto, if`
Subroutine	`JSUB m · RSUB`	`48 · 4C`	L ← PC, PC ← m · PC ← L	`call, return`
Input/output	`TD · RD · WD`	`E0 · D8 · DC`	test device, read byte, write byte	`getchar, putchar`

(m) means "the contents of memory at m". Students do not memorise opcodes: the assembler looks them up in its table (OPTAB, Lecture 3).

Book: Beck §1.3.1 and Appendix A

Assembler directives are your C declarations

A **directive** is an order to the assembler, not to the CPU. It produces **no instruction**: it only names memory and sets starting values, exactly like a declaration in C.

C declaration	SIC directive	Memory it reserves
`int five = 5;`	`FIVE  WORD  5`	one word, starting value 5
`int count;`	`COUNT  RESW  1`	one word, no starting value
`int table[100];`	`TABLE  RESW  100`	100 words = 300 bytes
`char buf[4096];`	`BUFFER  RESB  4096`	4096 bytes
`char eof[] = "EOF";`	`EOF  BYTE  C'EOF'`	3 bytes: 45 4F 46 (no \0 at the end!)
`char dev = 0xF1;`	`INPUT  BYTE  X'F1'`	1 byte given in hex
(program begins / ends)	`COPY START 1000 … END`	where the program is loaded, where it starts

Watch the orange row: a C string ends with an invisible `\0`. SIC's `BYTE C'EOF'` is exactly 3 bytes. Nothing marks the end, so the program must know the length.

Book: Beck §1.3.1 (directives used in §1.3.3 examples)

Program 1: if–else becomes compare and jump

C

```
if (numa > numb)
    max = numa;
else
    max = numb;
```

SAME AS C

Two paths meet

Both paths end at one place (ABIG), just like the code after an if–else in C.

SIC (our assembler's listing)

```
addr  label  instr  operand  code
1000         LDA    NUMA     001012
1003         COMP   NUMB     281015
1006         JGT    ABIG     34100C
1009         LDA    NUMB     001015
100C  ABIG   STA    MAX      0C1018
...
1012  NUMA   WORD   17
1015  NUMB   WORD   42
1018  MAX    RESW   1
```

LDA NUMA

A ← numa, as if "then" were already chosen.

COMP NUMB

Compare A with numb: 17 < 42, so CC = '<'.

JGT ABIG

Jump only if CC is '>'. It is not, so carry on.

LDA NUMB

The else part: replace A with numb (42).

STA MAX

Both paths meet here. max = 42.

Different from C: the > of C is split in two. **COMP** only remembers the result in the condition code; the **jump** reads it later. Notice the trick too: A already holds numa, so the "then" part needs no instruction at all.

Book: Beck §1.3.3 (SIC programming examples)

Program 2: a for-loop with the index register

C

```c
for (i = 0; i < 5; i++)
    str2[i] = str1[i];
```

SIC

```
        LDX   ZERO
MOVECH  LDCH  STR1,X
        STCH  STR2,X
        TIX   FIVE
        JLT   MOVECH
...
STR1    BYTE  C'HELLO'
STR2    RESB  5
ZERO    WORD  0
FIVE    WORD  5
```

C `i = 0` — SIC `LDX ZERO`

X starts at 0. X is our i.

C `str2[i] = str1[i]` — SIC `LDCH STR1,X · STCH STR2,X`

Copy one byte through A. Indexed: TA = STR1 + X, then STR2 + X.

C `i++ and i < 5` — SIC `TIX FIVE`

Two jobs in one: X = X + 1, then compare X with 5 and set CC.

C `back to the top` — SIC `JLT MOVECH`

If CC is '<', jump to MOVECH. When X reaches 5, CC is '=' and we fall through.

*Run it: **21 instructions** (1 + 5 rounds × 4) copy HELLO. The next slide shows every one.*

Watch the loop run, instruction by instruction

A C loop, running as SIC instructions

C (what you wrote in 1st semester)

```
for (i = 0; i < 5; i++)
    str2[i] = str1[i];
```

```
1000         LDX   ZERO
1003  MOVECH  LDCH  STR1,X
1006         STCH  STR2,X
1009         TIX   FIVE
100C         JLT   MOVECH
```

Registers

X	A	CC
0	—	—
index (like i)	accumulator	condition code

	[0]	[1]	[2]	[3]	[4]
STR1	H	E	L	L	0
	1012	1013	1014	1015	1016
STR2					
	1017	1018	1019	101A	101B

The same loop twice: C on top, the SIC instructions the assembler would produce below. Watch X: it plays the role of i.

Click the video to play or pause. About 1 minute, no sound.

Watch for

1. The **orange C part** at the top-left changes as each SIC line runs.
2. **X** grows by 1 only at TIX.
3. **A** holds one character at a time: the byte travels through it.
4. The **target address** in the caption is always STR + X.
5. In round 5, CC becomes **'='** and JLT stops jumping.

Program 3: a function call with JSUB and RSUB

C

```c
int square(int x)
{
    return x * x;
}

res1 = square(7);
res2 = square(12);
```

SAME AS C

Call and return

Like C: jump into the function, run it, come back to the line after the call.

SIC

```
1000         LDA   NUM1
1003         JSUB  SQUARE
1006         STA   RES1
1009         LDA   NUM2
100C         JSUB  SQUARE
100F         STA   RES2
...
1015  SQUARE LDA   TEMP
1018         MUL   TEMP
101B         RSUB
```

JSUB at 1003
L ← 1006 (the next instruction), PC ← 1015

RSUB
PC ← L = 1006. A = 49, stored in RES1

JSUB at 100C
L ← 100F, PC ← 1015

RSUB
PC ← L = 100F. A = 144, stored in RES2

Different from C: the argument and the result both travel in **A**, by agreement only. And the return address lives in **L**, not on a stack: if SQUARE called another routine, it would first have to save L with STL.

Book: Beck §1.3.1 (JSUB, RSUB, register L)

What SIC/XE adds to SIC

	SIC	SIC/XE
Memory	32 KB (2^{15} bytes)	1 MB (2^{20} bytes)
Registers	A, X, L, PC, SW	+ B (base), S, T (general), F (48-bit float)
Instruction formats	one: 3 bytes	four: 1, 2, 3 or 4 bytes
Addressing modes	direct, indexed	+ immediate, indirect, PC-relative, base-relative
Floating point	none	48-bit (ADDF, MULF …)
Register-to-register	none	ADDR S,A · COMPR A,S · CLEAR X …

Why the changes?

1 MB of memory needs 20-bit addresses, which do not fit in the old 15-bit field. So XE adds a 4-byte format.

More registers mean fewer trips to memory. In C terms: more values can stay "in your hands" instead of in variables.

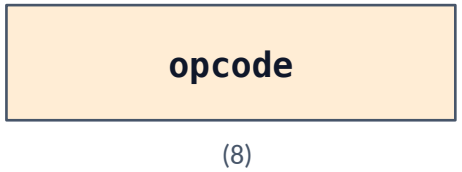
XE is **upward compatible**: an old SIC program still runs.

Book: Beck §1.3.2

The four instruction formats of SIC/XE

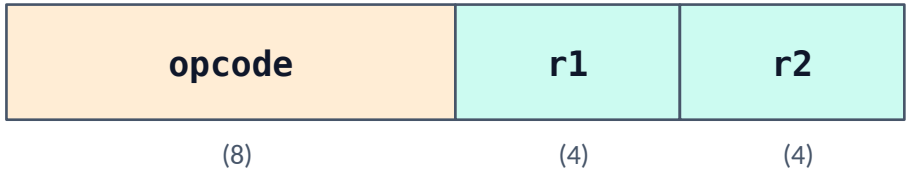
Format 1 · 1 byte

no operand, e.g. FIX, HIO



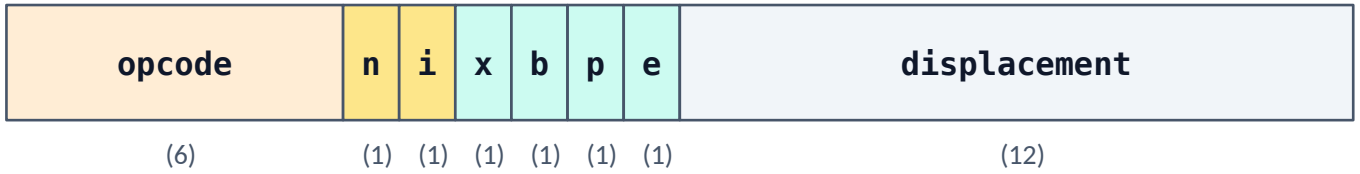
Format 2 · 2 bytes

two registers: ADDR S,A = 9040



Format 3 · 3 bytes

the common case: LDA ALPHA



Format 4 · 4 bytes

e = 1, written with a +: +JSUB RDREC



n, i say how to **use** the target address (simple, immediate, indirect). **x, b, p** say how to **compute** it (indexed, base, PC). **e** says format 3 or 4.

Book: Beck §1.3.2 (instruction formats)

Addressing modes, explained with C

Mode	Flag bits	SIC/XE	Rule	C analogy	In words
Simple	`n=1, i=1`	`LDA ALPHA`	`operand = (TA)`	`a = alpha;`	the usual case
Immediate	`n=0, i=1`	`LDA #3`	`operand = TA itself`	`a = 3;`	a constant inside the instruction
Indirect	`n=1, i=0`	`LDA @P`	`operand = ((TA))`	`a = *p;`	TA holds the address of the operand
Indexed	`x=1`	`LDA TABLE,X`	`TA = disp + (X)`	`a = table[i];`	X is the index
Base-relative	`b=1`	`LDA FIELD`	`TA = (B) + disp`	`a = s->field;`	base pointer + fixed offset
PC-relative	`p=1`	`LDA ALPHA`	`TA = (PC) + disp`	`(no C syntax)`	"600 bytes after this instruction"

SAME AS C

Immediate, indirect, indexed

are C's literal, *p and a[i]. You already use them every day.

DIFFERENT FROM C

PC-relative

has no C syntax. The assembler picks it; the program then works at any load address.

Book: Beck §1.3.2 (addressing modes)

Worked examples: six LDA instructions, six modes

Given: **(B) = 006000**, **(PC) = 003000** (already moved past the instruction), **(X) = 000090**.

Address	Contents
3030	003600
3600	103000
6390	00C303
C303	003030

Cover the orange columns, decode each row, then click to check.

All LDA (opcode 00). Values checked with our SIC/XE decoder.

Machine code	n i x b p e	Mode	Target address	A after LDA
032600	1 1 0 0 1 0	PC-relative	3000 + 600 = 3600	**103000**
03C300	1 1 1 1 0 0	base + indexed	6000 + 300 + 90 = 6390	**00C303**
022030	1 0 0 0 1 0	indirect, PC-rel.	3000 + 30 = 3030 → 3600	**103000**
010030	0 1 0 0 0 0	immediate	30	**000030**
003600	0 0 (SIC)	old SIC format	3600	**103000**
0310C303	1 1 0 0 0 1	format 4	C303	**003030**

Book: after the addressing-mode example in Beck §1.3.2

SIC/XE and x86-64: same ideas, bigger machine

	SIC/XE (on paper)	x86-64 (lab PC)
Word size	3 bytes	8 bytes (registers), int = 4 bytes
Registers	A, X, L, B, S, T, F, PC, SW	16 general (rax … r15), rip, rflags
Instruction length	1 to 4 bytes	1 to 15 bytes
Condition code	in SW: <, =, >	flags in rflags: ZF, SF, CF, OF
Function call	JSUB puts the return address in L	call pushes it on the stack; ret pops it
Addressing	simple, #, @, ,X, base, PC-relative	the same ideas: $5, (%rax), 8(%rbp), (%rip)

Two spellings of the same CPU

AT&T (gcc output): `movl $0, %eax`
source first, then destination; `%` on registers, `$` on constants.

Intel (nasm, our labs): `mov eax, 0`
destination first, like C's `eax = 0;`

Good news for C programmers

x86-64 has a real **stack** with push, pop, call and ret. So functions, local variables and recursion work in hardware the way you know from C, unlike SIC's single L register.

Book: CS:APP §3.2–3.4, §3.7 (procedures and the stack)

Your C, compiled: if–else in x86-64

maxab.c

```c
int max(int a, int b)
{
    if (a > b)
        return a;
    else
        return b;
}
```

Same pattern as Program 1 on SIC: compare, jump, two paths meeting at one label.

gcc -O0 -S maxab.c (AT&T)

```
max:
    pushq   %rbp
    movq    %rsp, %rbp
    movl    %edi, -4(%rbp)
    movl    %esi, -8(%rbp)
    movl    -4(%rbp), %eax
    cmpl    -8(%rbp), %eax
    jle     .L2
    movl    -4(%rbp), %eax
    jmp     .L3
.L2:
    movl    -8(%rbp), %eax
.L3:
    popq    %rbp
    ret
```

Parameters arrive in registers

a in **edi**, b in **esi**: the calling convention. gcc -O0 first copies them into the stack frame at -4 and -8 from rbp, like C local variables.

cmpl, then jle

Compare, then jump: exactly SIC's COMP and JGT. But gcc jumps on the **opposite** test, a ≤ b, straight to the else part (.L2).

Labels .L2, .L3

The if–else became **two labels and two jumps**. The result goes in **eax** for the return.

Book: CS:APP §3.6 (control)

Key ideas from Class 2

1. A SIC instruction is 24 bits: an 8-bit opcode, the x bit and a 15-bit address.
2. if-else becomes COMP and a conditional jump; a for-loop uses the index register X and TIX.
3. JSUB saves the return address in L; RSUB jumps back to it.
4. SIC/XE adds registers, four instruction formats and addressing modes (immediate #, indirect @, PC- and base-relative).
5. x86-64 uses the same ideas: compare, jump, call and return, only with more registers.

Homework and practice

Homework 2 (bring it to the next class)

1. Encode LDA ALPHA by hand (ALPHA = 1033) and STCH BUFFER,X (BUFFER = 1039). Show the bits.
2. Trace Program 2 with C'CAT' and THREE: write X, A and CC after every instruction.
3. Name the addressing mode of LDA #3, LDA @PTR and LDA VALUE in SIC/XE, and what each loads.

Try it in the browser

SIC simulator: step through the slide programs

`labs/lab02.html`

On the course website: open the lab page and choose "Try it here".

Next class: Assemblers